

Introducción al aprendizaje de máquina

NPL-LLMs

2025

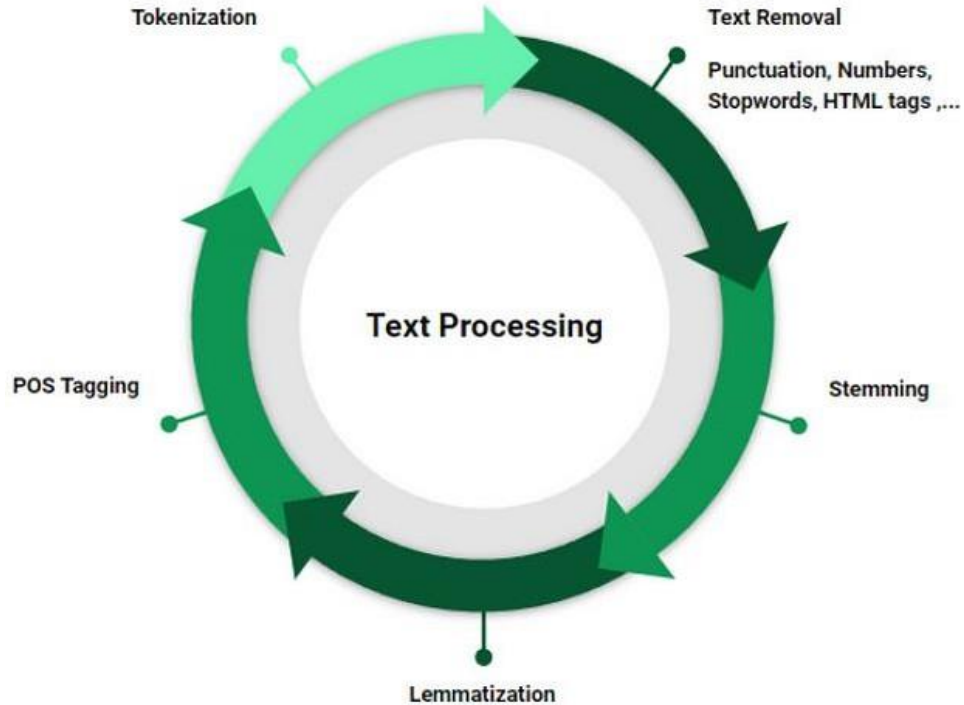
Yomin Jaramillo M
yejaramilm@eafit.edu.co

Agenda

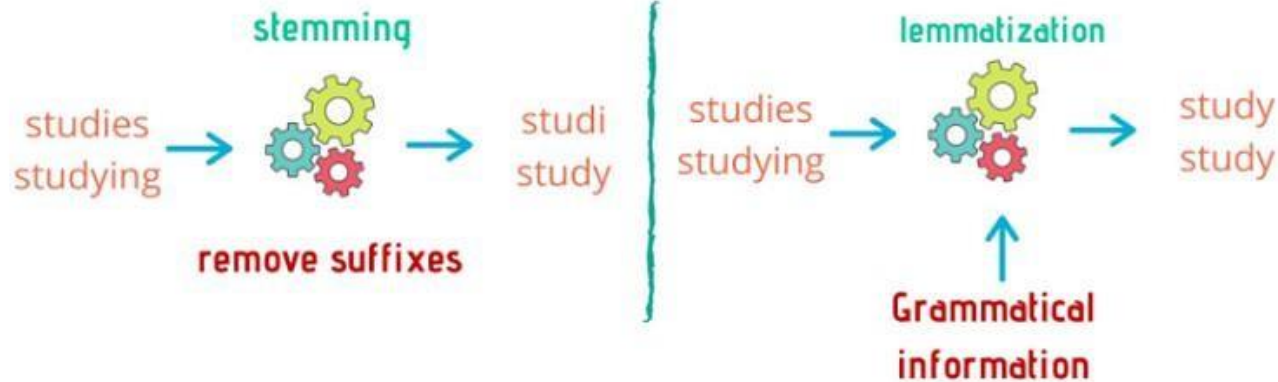


- Introducción a NLP
- Aplicaciones de NLP
- NLP con transformers
- Ejemplo práctico de Q&A usando Hugging Face.

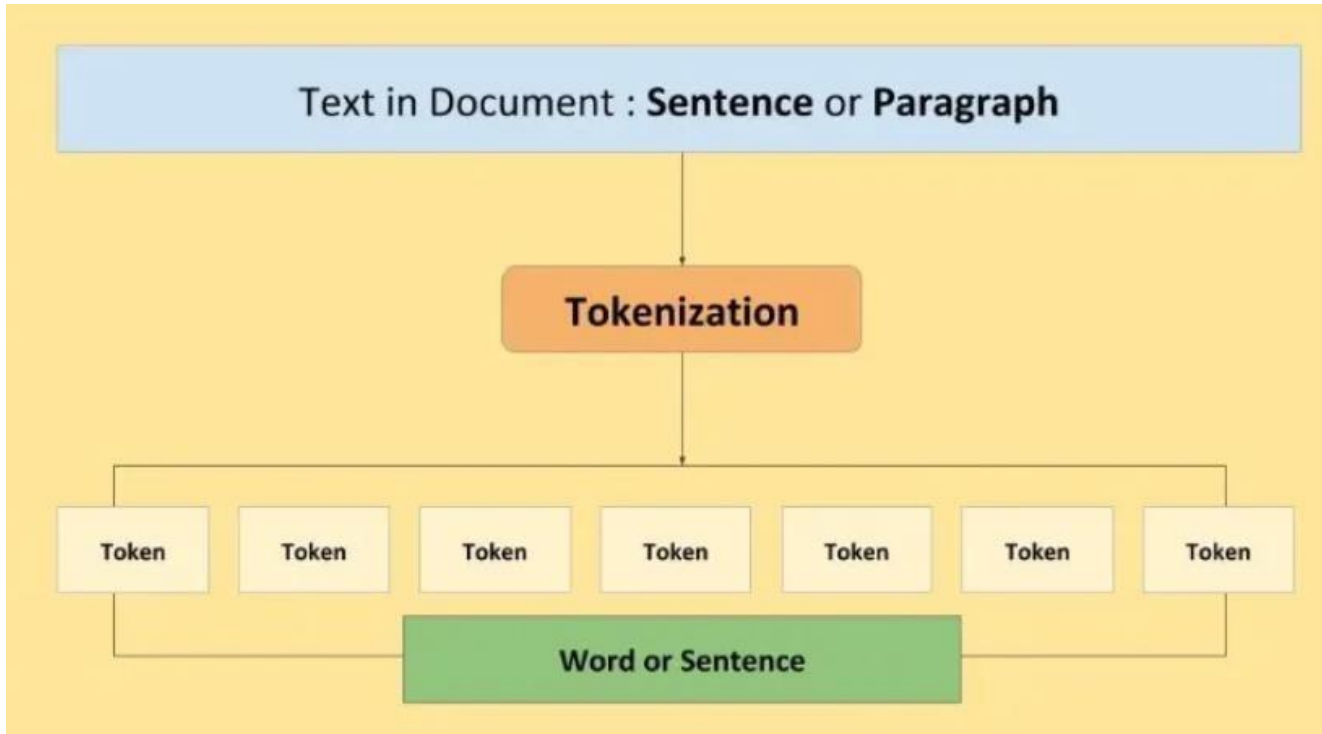
Introducción a NLP



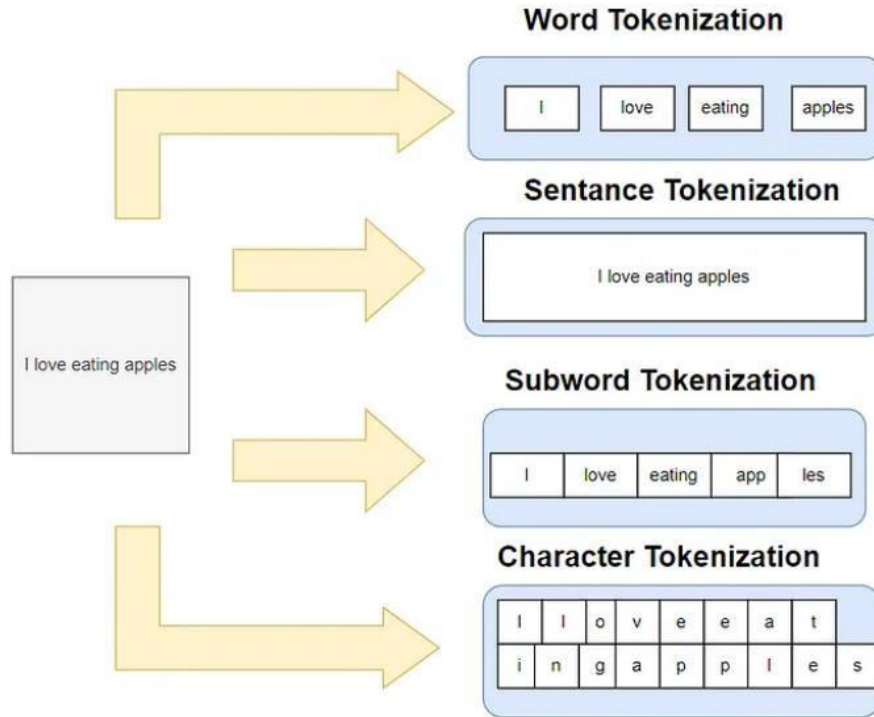
Stemming and Lemmatization



Tokenization

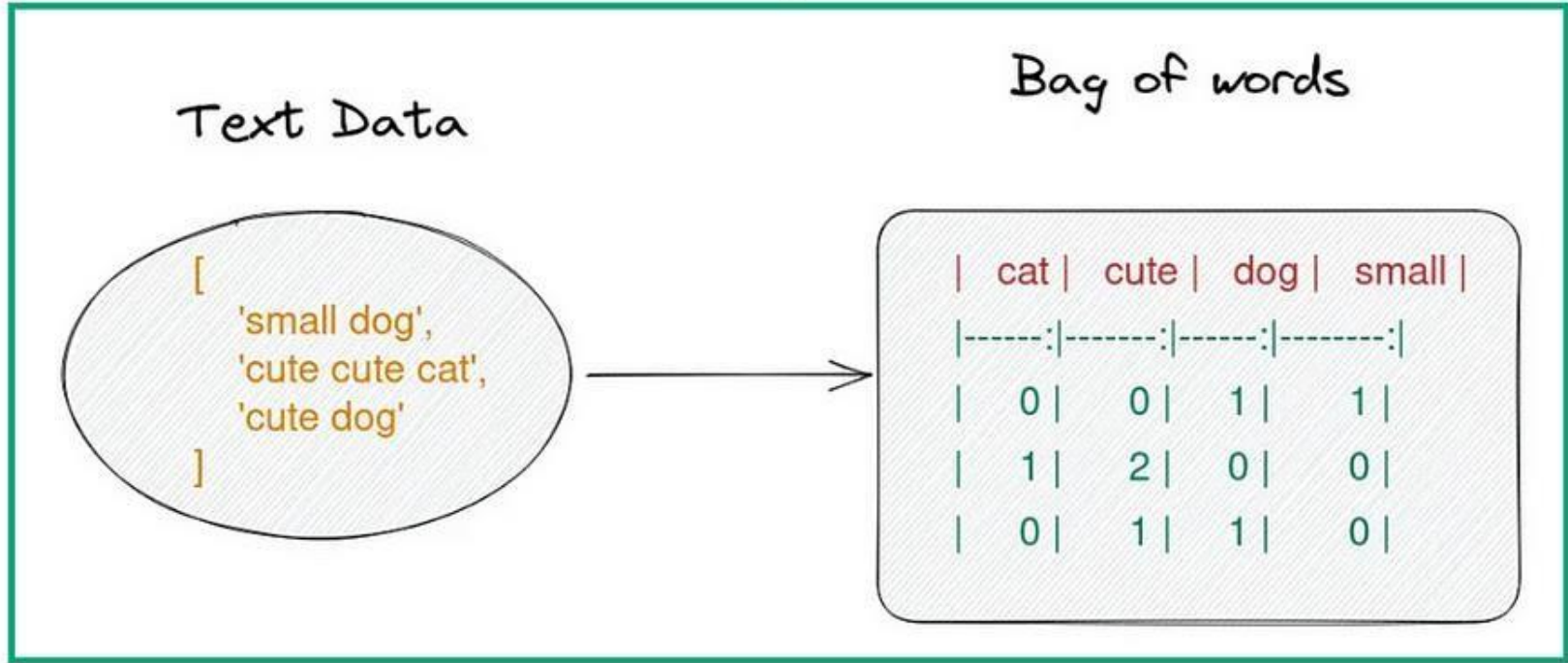


Tokenization

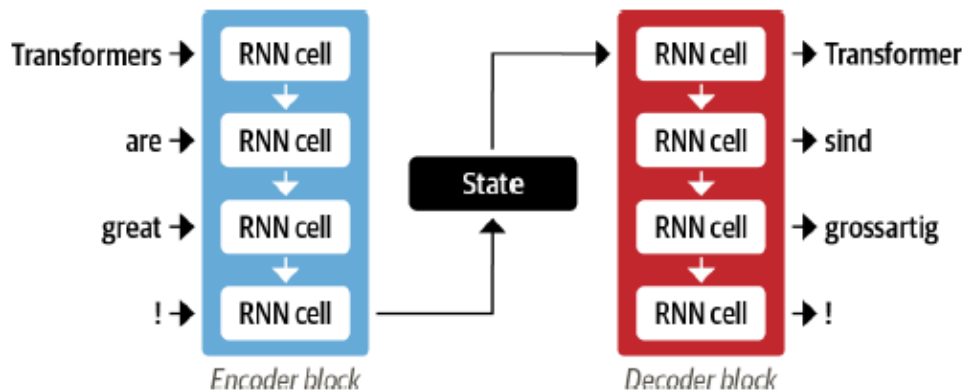
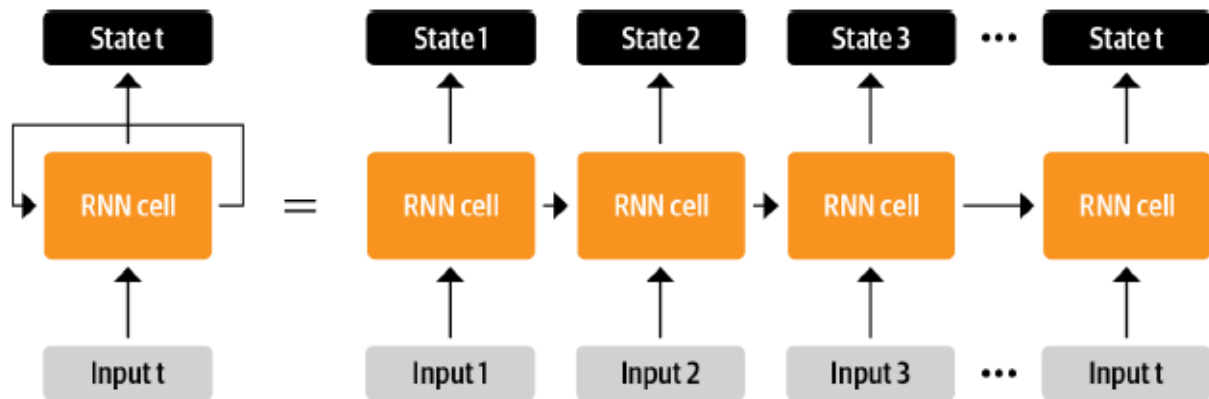


Tokenization techniques

Bag of words



RNN



Hello Transformers

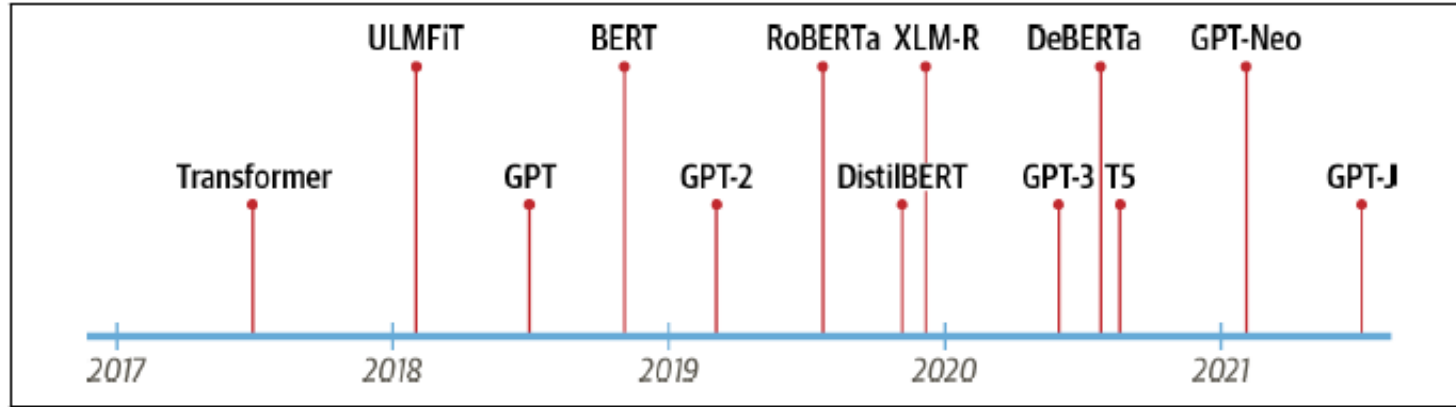
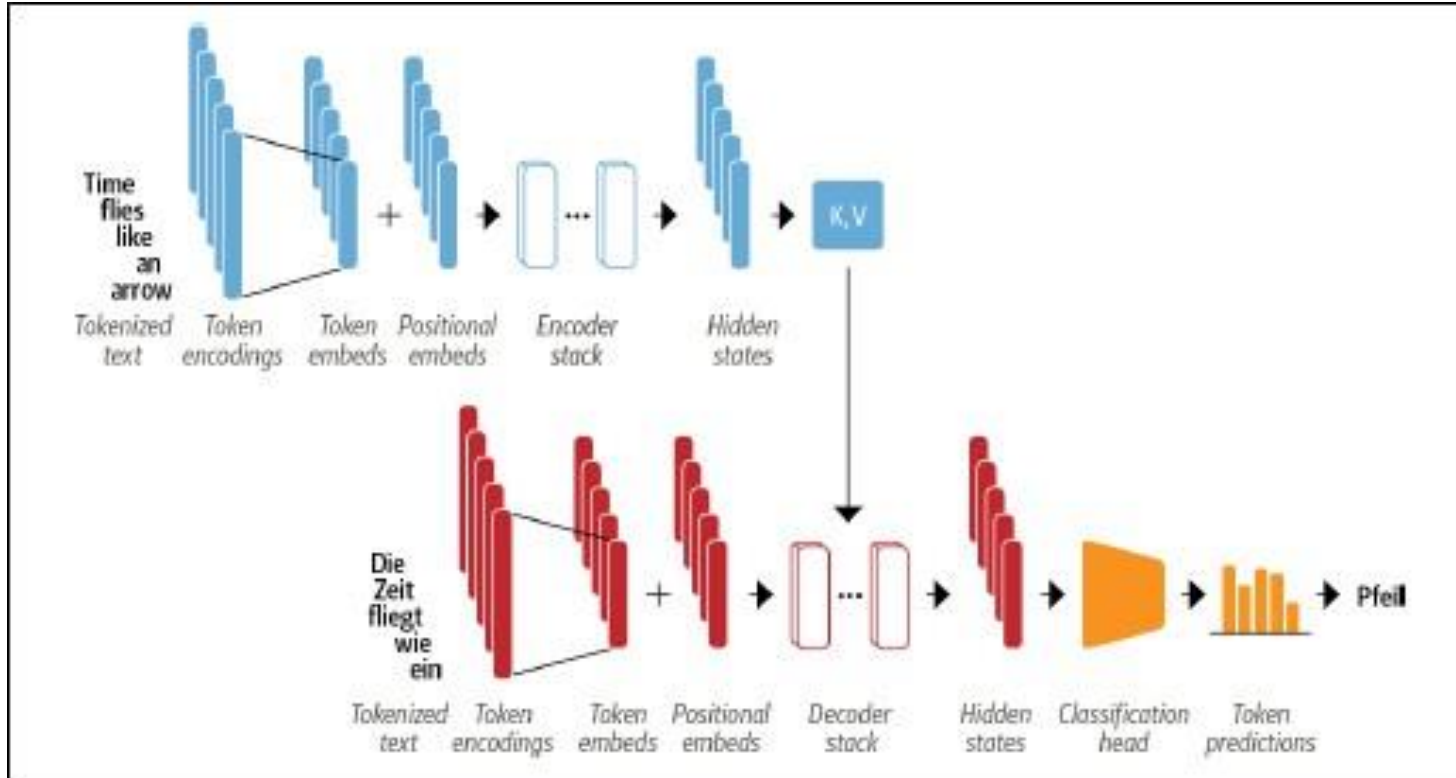
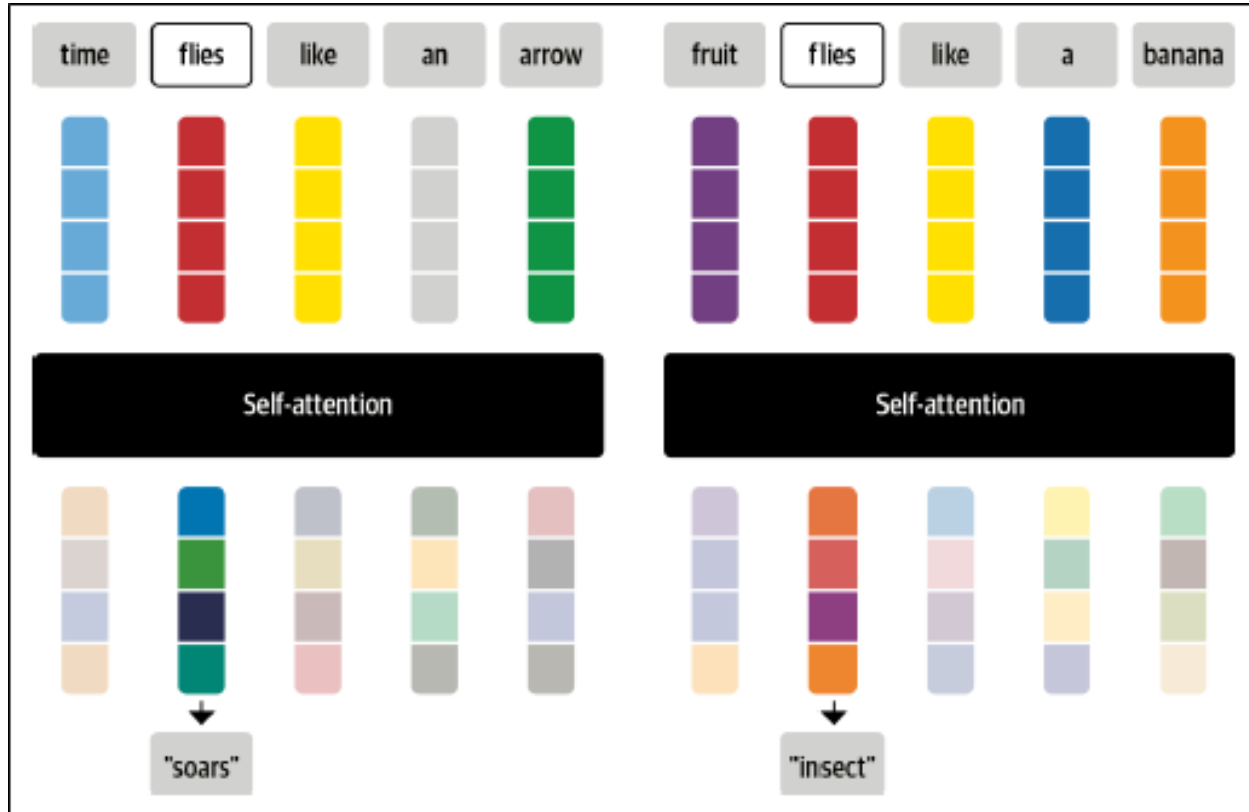


Figure 1-1. The transformers timeline

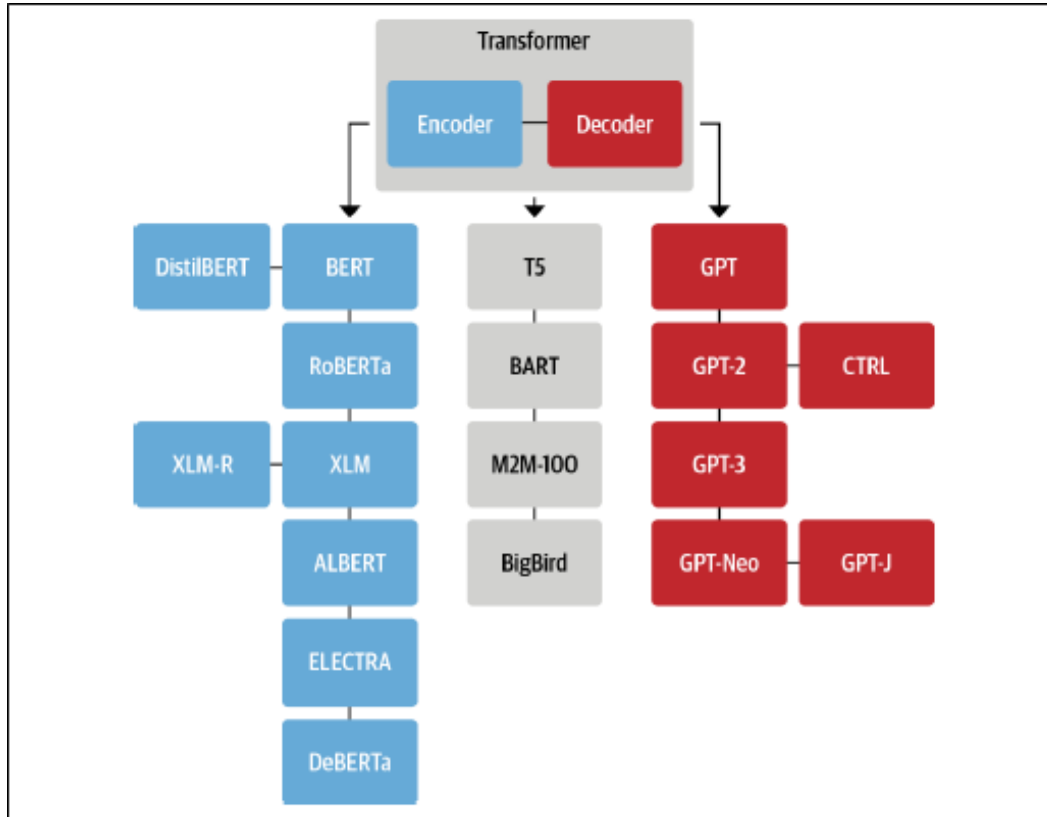
Transformer Architecture



Self-Attention



Algunas Arquitecturas de Transformers



Hugging Face Hub



Hugging Face

Search models, datasets, users...

Models

Datasets

Spaces

Docs

Solutions

Pricing



Log In

Sign Up

Tasks Libraries Datasets Languages Licenses Other

Filter Tasks by name

Multimodal

- Feature Extraction
- Text-to-Image
- Image-to-Text
- Text-to-Video
- Visual Question Answering
- Document Question Answering
- Graph Machine Learning

Computer Vision

- Depth Estimation
- Image Classification
- Object Detection
- Image Segmentation
- Image-to-Image
- Unconditional Image Generation
- Video Classification
- Zero-Shot Image Classification

Natural Language Processing

- Text Classification
- Token Classification
- Table Question Answering
- Question Answering
- Zero-Shot Classification
- Translation
- Summarization
- Conversational
- Text Generation
- Text2Text Generation
- Fill-Mask
- Sentence Similarity

Models 372,969

Filter by name

new Full-text search

Sort: Trending

adept/fuyu-8b

Text Generation • Updated 4 days ago • 14.2k • 508

segmind/SSD-1B

Text-to-Image • Updated about 8 hours ago • 8.62k • 175

mistralai/Mistral-7B-v0.1

Text Generation • Updated 14 days ago • 277k • 1.48k

stabilityai/stable-diffusion-xl-base-1.0

Text-to-Image • Updated 24 days ago • 7.37M • 3.28k

meta-llama/Llama-2-7b-chat-hf

Text Generation • Updated 4 days ago • 975k • 1.55k

CausalLM/7B

Text Generation • Updated 1 day ago • 184 • 74

meta-llama/Llama-2-7b

Text Generation • Updated Jul 19 • 2.86k

SimianLuo/LCM_Dreamshaper_v7

Text-to-Image • Updated 1 day ago • 24.1k • 78

jinaai/jina-embeddings-v2-base-en

Feature Extraction • Updated about 3 hours ago • 4.17k • 183

HuggingFaceH4/zephyr-7b-alpha

Text Generation • Updated about 11 hours ago • 58.6k • 799

CausalLM/14B

Text Generation • Updated about 5 hours ago • 323 • 136

amazon/MistralLite

Text Generation • Updated 3 days ago • 2.94k • 122

mistralai/Mistral-7B-Instruct-v0.1

Text Generation • Updated 15 days ago • 220k • 864

teknium/OpenHezmes-2-Mistral-7B

Text Generation • Updated about 19 hours ago • 10.1k • 143

SkunkworksAI/BakLLaVA-1

Text Generation • Updated 3 days ago • 523 • 175

runwayml/stable-diffusion-v1-5

Text-to-Image • Updated Aug 23 • 7.65M • 9.47k

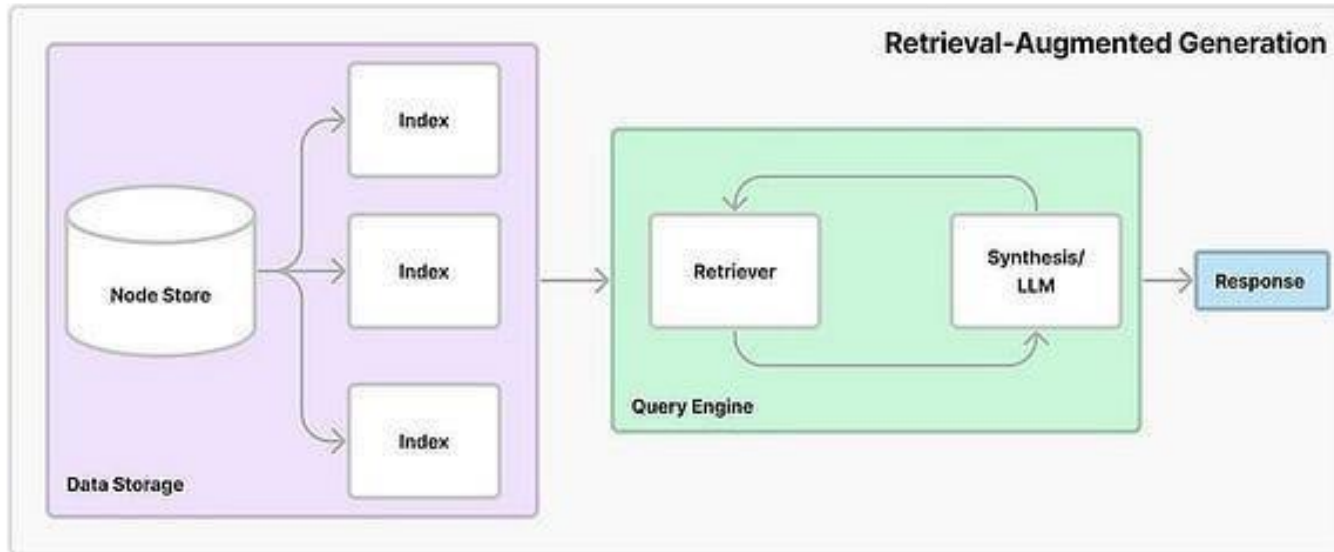
RAG

Retrieval-Augmented Generation (RAG)

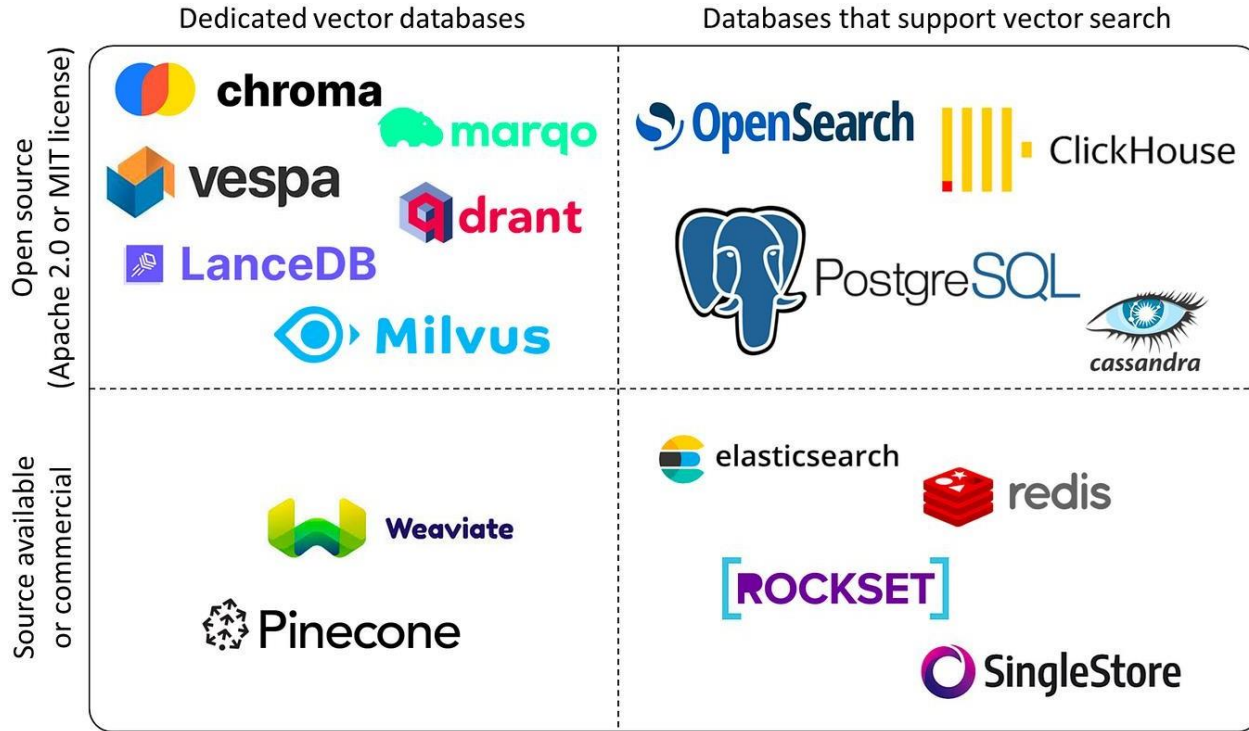
Retrieval-augmented generation (RAG) is a natural language processing (NLP) approach that combines the benefits of both retrieval-based and generation-based methods for content generation tasks. It aims to improve the quality and controllability of the generation tasks by leveraging a pre-trained language model in conjunction with a retrieval mechanism.



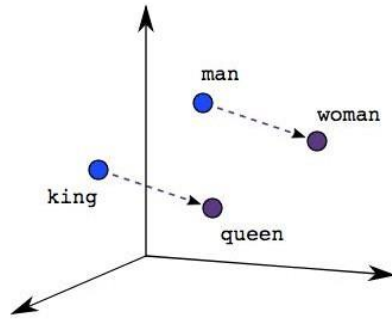
RAG: Componentes



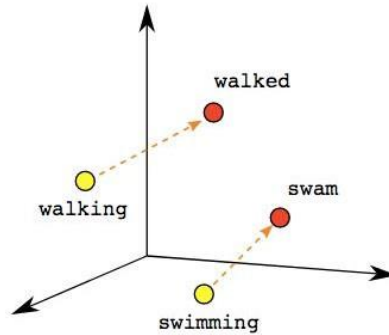
RAG: Database



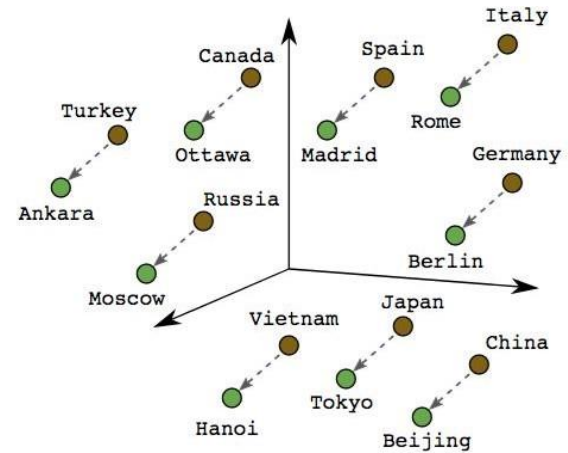
RAG: Embeddings



Male-Female



Verb Tense



Country-Capital

RAG: LLM



Hugging Face

The task illustrated in this tutorial is supported by the following model architectures:

ALBERT, BART, BERT, BigBird, BigBird-Pegasus, BLOOM, CamemBERT, CANINE, ConvBERT, Data2VecText, DeBERTa, DeBERTa-v2, DistilBERT, ELECTRA, ERNIE, ErnieM, Falcon, FlauBERT, FNet, Funnel Transformer, OpenAI GPT-2, GPT Neo, GPT NeoX, GPT-J, I-BERT, LayoutLMv2, LayoutLMv3, LED, LiLT, Longformer, LUKE, LXMERT, MarkupLM, mBART, MEGA, Megatron-BERT, MobileBERT, MPNet, MPT, MRA, MT5, MVP, Nezha, Nyströmformer, OPT, QDQBERT, Reformer, RemBERT, RoBERTa, RoBERTa-PreLayerNorm, RoCBert, RoFormer, Splinter, SqueezeBERT, T5, UMT5, XLM, XLM-RoBERTa, XLM-RoBERTa-XL, XLNet, X-MOD, YOSO